

AI CAREER INTELLIGENCE

REPORT 2026

Deep Research Edition | Evidence-Based | Fresher-Focused

PREPARED FOR

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CGPA: 8.84/10 | Target: FAANG/Product Company AI Roles

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EXECUTIVE SUMMARY

A no-nonsense overview of the AI career landscape in 2026 for Indian fresher engineers

40%+ YoY growth in AI job postings (NASSCOM)	■5–12L Fresher AI salary range in India	3% Engineering grads with real AI/ML skills	280% Agentic AI job growth YoY
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The Honest Situation in 2026

India produces 1.5 million engineering graduates per year. Fewer than 3% have real, deployable AI/ML skills. That gap is your single biggest opportunity. Companies are not looking for graduates who watched YouTube tutorials — they want engineers who have built, deployed, and maintained real AI systems. The barrier to entry has shifted from academic credentials to demonstrable work.

The AI job market in 2026 is NOT a monolith. A data annotation contractor earns ■3 LPA. A senior LLM engineer at a product company earns ■70 LPA. Both are 'AI jobs.' This report helps you understand exactly where to position yourself to land in the higher end of that spectrum — within 6 months of serious preparation.

5 Brutal Truths About AI Careers in 2026

Truth: Notebooks are not portfolios.

If your projects exist only in Jupyter notebooks and have never served a real request, they do not differentiate you. Recruiters have seen 10,000 Titanic classification notebooks.

Truth: Generative AI is real, but saturated at entry-level.

Everyone is claiming 'GenAI experience.' The differentiation is in RAG pipelines, fine-tuning, and agentic systems — not just calling the OpenAI API.

Truth: IT services vs product companies is a 60-150% salary difference.

Joining TCS/Infosys as an AI fresher means ■6-8 LPA. Joining a product company or GCC with the same skills means ■10-16 LPA. The effort to aim higher is worth it.

Truth: DSA still matters, even for AI roles.

Google, Amazon, Microsoft, and Indian unicorns all gate AI roles behind DSA rounds. You cannot skip this. But you don't need LeetCode Hard — LeetCode Medium competence is the realistic target.

Truth: Layoffs are real, but they're hiring simultaneously.

Microsoft laid off 15,000+ employees in 2025 while spending \$80B on AI infrastructure. The same companies reducing headcount in non-AI roles are desperately hiring AI specialists. Specialisation is your insurance policy.

Bottom Line for Marco: You are a 3rd-year CSE (AI/ML) student with strong projects, leadership, and a 8.84 CGPA. You have competitive advantages over most of your peers. The gap between you and a ■18-25 LPA offer is 6 months of disciplined execution on the right things — not motivation, but a concrete plan. This report gives you that plan.

SECTION 1: AI INDUSTRY DOMAIN BREAKDOWN

What domains are real, what is hype, and where freshers actually get hired

1.1 The 12 Major AI Domains — Reality Check

Not all AI domains are equal in terms of hiring demand, salary, or fresher accessibility. Below is a comprehensive breakdown of every major AI domain with realistic assessments.

Domain	Growth	Hiring Demand	Fresher Access	6-Mo Ready?	India Salary (Fresher)
Generative AI / LLM Eng.	■ Explosive	Very High	Medium	Yes (with projects)	■8-15 LPA
Agentic AI	■ 280% YoY	Very High	Low-Medium	Possible	■10-18 LPA
MLOps / AI Engineering	■ Strong	High	Medium	Yes	■8-14 LPA
NLP / Text AI	■ Strong	High	Medium	Yes	■8-12 LPA
Computer Vision	■ Steady	Medium-High	Medium	Yes	■7-12 LPA
Applied AI (Backend+AI)	■ Strong	Very High	Medium-High	Yes	■8-16 LPA
ML Engineering	■ Strong	High	Medium	Yes	■7-12 LPA
Data Engineering for AI	■ Strong	High	High	Yes	■6-10 LPA
AI Research	■ Niche	Low	Very Low	No (2+ yrs)	■6-15 LPA
Reinforcement Learning	■ Niche	Low-Medium	Very Low	No	■10-20 LPA
Robotics AI	■ Emerging	Low (India)	Very Low	No	■8-14 LPA
Edge AI	■ Emerging	Low-Medium	Low	Unlikely	■8-12 LPA
Recommendation Systems	■ Steady	Medium	Low	Partially	■8-15 LPA
AI Security / Red Teaming	■ Growing	Low-Medium	Low	Unlikely	■10-18 LPA

1. Generative AI / LLM Engineering

What It Means:	Building, fine-tuning, and deploying large language models for products. Includes RAG pipelines, chatbots, document AI, code generation tools.
Key Companies:	OpenAI, Anthropic, Google DeepMind, Microsoft, Sarvam AI, Krutrim, Postman, Sprinklr, HaptikAI, virtually all Indian unicorns now.
Growth Signal:	GenAI job postings grew 800% for non-tech roles since ChatGPT launch (2022). Demand for GenAI skills showed a 4x increase in India (NASSCOM 2024).
Fresher Access:	HIGH — If you can build RAG pipelines, work with LangChain/LlamaIndex, fine-tune open-source models on HuggingFace, and deploy via FastAPI, you are hireable.
Automation Risk:	MEDIUM — Entry-level prompt engineering is getting commoditized. Deep LLM engineering (fine-tuning, RLHF, evaluation frameworks) remains specialized.

Salary (India):	Fresher: ■8-15 LPA Mid: ■20-40 LPA Senior: ■50-80 LPA+
Verdict:	■ BEST DOMAIN FOR FRESHER IN 6 MONTHS

2. Agentic AI

What It Means:	Building autonomous AI systems that plan, reason, use tools, and execute multi-step tasks. Uses frameworks like LangGraph, AutoGen, CrewAI, custom orchestration.
Key Companies:	Anthropic (scaling 5x in 2026), OpenAI (hiring 3,500+), Salesforce, ServiceNow, enterprise automation companies.
Growth Signal:	280% YoY job growth. Forward-deployed engineer listings grew 800% in 2025. The hottest specialization in 2026.
Fresher Access:	MEDIUM — Requires solid LLM knowledge first. Achievable in 6-8 months if you have strong Python and LLM foundation.
Automation Risk:	LOW — Agentic systems are complex engineering problems. Unlike basic chatbots, they require real software engineering skills that are hard to automate.
Salary (India):	Fresher: ■10-18 LPA Senior: ■50-100 LPA
Verdict:	■ STRONG SECOND FOCUS — BUILDS ON LLM FOUNDATION

3. MLOps / AI Engineering

What It Means:	Deploying, monitoring, maintaining, and scaling ML models in production. Involves Docker, Kubernetes, CI/CD, model registries, drift detection, A/B testing.
Key Companies:	All companies with production AI — Amazon, Google, Flipkart, Zomato, Swiggy, Meesho.
Growth Signal:	Strong and consistent. The 'MLOps bottleneck is real' — companies have invested in models but underinvested in production infrastructure.
Fresher Access:	MEDIUM — Requires backend/DevOps familiarity. Strong differentiation from pure ML candidates.
Automation Risk:	LOW — Production reliability is a hard engineering problem. MLOps engineers that keep AI systems alive affect revenue directly.
Salary (India):	Fresher: ■8-14 LPA Senior: ■45-60 LPA at product/GCC
Verdict:	■ EXCELLENT COMPLEMENTARY SKILL — ADDS HUGE VALUE TO ANY AI ROLE

4. AI Research

What It Means:	Publishing papers, advancing state-of-the-art in ML. Requires PhD-level math, deep theoretical knowledge, 2-3 years minimum of specialized study.
Key Companies:	DeepMind, OpenAI Research, Microsoft Research, IIT labs.
Growth Signal:	Niche. Fewer than 5,000 true AI research positions globally at any given time.
Fresher Access:	VERY LOW — Do not pursue if your goal is employment within 6-12 months. Requires PhD or exceptional published work.
Automation Risk:	HIGH over time — research roles are increasingly automated by AI itself at the exploratory level.
Salary (India):	Fresher/PhD: ■15-25 LPA in India US: \$150K-\$300K

Verdict:	■ WRONG PATH FOR YOUR TIMELINE — SKIP UNLESS YOU WANT A PHD
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5. Computer Vision

What It Means:	Training and deploying models that interpret images/video. Use cases: quality control, medical imaging, autonomous vehicles, security cameras, retail analytics.
Key Companies:	NVIDIA, Tesla, Qualcomm, Intel, healthcare companies, retail AI startups.
Growth Signal:	Steady. 36,000+ openings in US alone for Computer Vision Engineers (2025). Strong in India for surveillance, automotive, manufacturing.
Fresher Access:	MEDIUM — Achievable with PyTorch, CNN knowledge, and a solid project (e.g., object detection, face recognition, anomaly detection).
Automation Risk:	MEDIUM — The domain is well-established with many experienced practitioners. Differentiation requires real data and deployment experience.
Salary (India):	Fresher: ■7-12 LPA Senior: ■25-40 LPA
Verdict:	■ VIABLE — ESPECIALLY IF COMBINED WITH EDGE AI OR HEALTHCARE AI

6. Reinforcement Learning

What It Means:	Training agents to make sequential decisions through reward signals. Used in robotics, game AI, trading, recommendation optimization, RLHF for LLMs.
Key Companies:	DeepMind, OpenAI, Tesla, quantitative trading firms, Meta AI.
Growth Signal:	Growing but niche. The RLHF (RL from Human Feedback) component is critical for LLM training.
Fresher Access:	VERY LOW — Extremely math-heavy. Requires strong probability, linear algebra, dynamic programming. Not beginner-friendly.
Automation Risk:	LOW — True RL engineers are rare globally. But the barrier to entry is very high.
Salary (India):	Fresher: ■10-20 LPA Senior: ■50-100 LPA+
Verdict:	■■ AVOID AS PRIMARY FOCUS — PURSUE ONLY IF YOU HAVE STRONG MATH BACKGROUND

1.2 Domain Rankings Summary

Based on hiring demand, fresher accessibility, salary potential, and 6-month achievability:

Rank	Domain	Hiring	Fresher Ease	Salary Ceiling	6-Mo Target?
#1	Generative AI / LLM Engineering	★★★★★	★★★★■	■80 LPA+	YES
#2	Agentic AI	★★★★★	★★★██	■100 LPA+	YES (partial)
#3	MLOps / AI Engineering	★★★★■	★★★██	■60 LPA	YES
#4	Applied AI (Backend+AI)	★★★★■	★★★★■	■50 LPA	YES
#5	NLP / Text AI	★★★★■	★★★★■	■40 LPA	YES
#6	Computer Vision	★★★██	★★★██	■40 LPA	YES
#7	Data Engineering for AI	★★★★■	★★★★★	■35 LPA	YES
#8	Recommendation Systems	★★★██	★★███	■45 LPA	Partially

Rank	Domain	Hiring	Fresher Ease	Salary Ceiling	6-Mo Target?
#9	AI Security	★★■■■	★★■■■	■50 LPA	No
#10	Reinforcement Learning	★★■■■	★■■■■	■100 LPA+	No
#11	AI Research	★■■■■	★■■■■	■80 LPA+	No
#12	Robotics AI	★★■■■	★■■■■	■40 LPA	No

SECTION 2: LAYOFFS vs. HIRING REALITY

What actually happened, why, and what it means for your career strategy

2.1 The Layoff Timeline — 2023 to 2026

Understanding the layoff cycle is crucial to not panicking and making the right career moves. The story is more nuanced than 'AI is taking jobs.'

2022-23	Post-pandemic correction. Companies like Meta, Google, Amazon had 2x'd headcount during COVID boom. ~200,000 layoffs industry-wide. Primary reason: over-hiring correction, NOT AI.
2024	Transitional phase. Layoffs declined from 2023 peaks. Companies began restructuring towards AI investment priorities. Amazon's cuts framed as 'reallocation to AI infrastructure.'
2025	127,000+ tech workers laid off in US. AI explicitly cited in <8% of announcements (Crunchbase). Microsoft alone cut ~15,000 while spending \$80B on AI. Meta cut 10% while increasing AI capex.
2026	150,000+ cuts YTD. AI cited as factor in 25%+ of announcements. Simultaneously, AI role hiring is at all-time highs. The split is now visible: non-AI roles shrinking, AI roles expanding.

Key Insight: Bloomberg analysis suggests roughly half of 'AI-attributed' layoffs will result in the same roles being rehired offshore or at lower salaries. This is a labour repricing story as much as a displacement story. The real threat is to *generalist* software engineers — not to AI specialists.

2.2 Major Company Analysis

■ Google / Alphabet

AI Strategy	Gemini models, Google Cloud AI, DeepMind integration. Spending massive capex on AI infrastructure.
Layoff Reality	~12,000 in 2023 (post-pandemic). Additional cuts in 2025 in sales, recruiting, lower-priority engineering teams.
AI Hiring	Actively hiring for Gemini, Google Cloud AI, Applied ML, and DeepMind. GCC in Hyderabad and Bangalore is a major hiring hub.
Fresher Path	Extremely selective. Requires very strong DSA (LeetCode Hard level). ML roles require strong Python + ML fundamentals. Google STEP/GSOC for students.
Key Skills	Strong DSA, Python, ML fundamentals, system design, mathematics for ML
Verdict	Hard target. DSA is non-negotiable. Aim for internship first.

■ Microsoft

AI Strategy	Azure OpenAI, Copilot across all products, GitHub Copilot. \$80B AI infrastructure spend in 2025.
Layoff Reality	15,000 in 2025 (May + July). CEO Nadella confirmed AI writes 20-30% of Microsoft's code.
AI Hiring	Heavy hiring for Azure AI, Copilot teams, AI platform engineering. Microsoft India GCC in Hyderabad is top 3 employer for AI roles.

Fresher Path	Strong DSA needed. Also values system design for distributed AI systems. Campus hiring via MSIDC Hyderabad.
Key Skills	Python, Azure, distributed systems, ML fundamentals, strong coding
Verdict	Strong target for Hyderabad students. Apply for Azure ML and Copilot teams.

■ Amazon / AWS

AI Strategy	Bedrock (LLM platform), Alexa AI, AWS AI services, Amazon's internal AI for logistics/recommendations.
Layoff Reality	14,000 in Oct 2025 framed as reallocation to AI. Ongoing restructuring.
AI Hiring	Massive hiring for AWS ML/AI engineering, Amazon Alexa, robotics AI, recommendation systems.
Fresher Path	DSA-heavy hiring (LeetCode Medium is minimum). Leadership Principles interview is also critical.
Key Skills	Python, AWS, ML, distributed systems, DSA, leadership principles
Verdict	Very strong target. Amazon hires freshers in India. Focus on DSA + AWS certifications.

■ Meta

AI Strategy	Llama open source, Meta AI assistant, PyTorch, AI for ads/ranking, AR/VR AI.
Layoff Reality	10% cuts in early 2026 while accelerating AI capex. CEO Zuckerberg said AI agents will replace mid-level engineers.
AI Hiring	Hiring heavily for PyTorch, Llama, ranking systems, computer vision.
Fresher Path	Very selective. Strong research background valued. Open source contributions to PyTorch a huge plus.
Key Skills	PyTorch, large-scale ML, distributed training, research mindset
Verdict	Difficult. Build PyTorch open-source presence. Meta internship is the realistic path.

■ NVIDIA

AI Strategy	CUDA ecosystem, AI inference, NIM microservices, AI for everything. Stock +400% since 2022.
Layoff Reality	Minimal. Actually one of the biggest NET hirers in tech.
AI Hiring	CUDA engineers, inference optimization, AI software (NIM), partnerships team.
Fresher Path	Requires deep systems knowledge. CUDA, C++, GPU architecture. Not typical fresher path.
Key Skills	CUDA, C++, distributed computing, ML systems, performance optimization
Verdict	Hard path but incredible if achieved. Focus on CUDA projects if pursuing this.

■ Flipkart

AI Strategy	Recommendation systems, demand forecasting, logistics AI, GenAI for search.
Layoff Reality	Some restructuring in 2024 but stable overall. Hiring for AI/ML roles.
AI Hiring	Strong hiring for ML engineers, data scientists, AI platform engineers. Bangalore HQ.

Fresher Path	More accessible than FAANG. Values projects + DSA. Interview: DSA + ML + system design.
Key Skills	Python, ML, recommendation systems, A/B testing, SQL
Verdict	Strong target. 6-month prep achievable. Apply aggressively.

■ Zomato / Blinkit

AI Strategy	Delivery optimization, demand forecasting, restaurant recommendation AI, chatbots.
Layoff Reality	Minor restructuring. Generally growing.
AI Hiring	ML engineers, data scientists, AI product roles.
Fresher Path	Accessible. Strong DSA + Python + SQL + one good ML project can get you in.
Key Skills	Python, SQL, ML, operations research, optimization
Verdict	Realistic target. ■12-20 LPA for strong freshers.

■ PhonePe / Razorpay

AI Strategy	Fraud detection, credit scoring AI, transaction anomaly detection, customer service AI.
Layoff Reality	Growing. Razorpay expanded significantly in 2024-25.
AI Hiring	ML engineers for fraud/risk, data scientists, MLOps engineers.
Fresher Path	Moderate difficulty. Fintech AI roles value Python + ML + some SQL analytics.
Key Skills	Python, ML, anomaly detection, time-series analysis, SQL
Verdict	Strong fintech AI opportunity. ■12-22 LPA for skilled freshers.

■ Sarvam AI / Krutrim

AI Strategy	Indian language AI — building foundational models for Indic languages. Core AI research + product.
Layoff Reality	Growing startups. Net hirers.
AI Hiring	LLM engineers, data engineers for Indic data, ML researchers.
Fresher Path	Startup culture. Values projects heavily over degrees. Strong LLM knowledge + fast execution.
Key Skills	Python, LLMs, HuggingFace, fine-tuning, Indian language data
Verdict	Exciting opportunity. Equity upside. High risk, high reward.

■ Swiggy / Meesho

AI Strategy	Logistics optimization, demand prediction, personalization, supply chain AI.
Layoff Reality	Some in 2023-24 but stabilized. Meesho is profitable and hiring.
AI Hiring	ML engineers, data scientists, applied AI engineers.
Fresher Path	Moderate difficulty. Good pathway for strong freshers.
Key Skills	Python, ML, optimization, SQL, experimentation/A/B testing
Verdict	Solid mid-tier target. ■12-22 LPA range.

2.3 Which Jobs Are Most Threatened by AI

Role	Threat Level	Reason
Junior QA / Manual Testing	HIGH	AI testing tools automate routine test case generation.
Basic Data Entry / Annotation	VERY HIGH	Largely automated. Only quality control roles remain.
Junior Reporting / BI Analyst	HIGH	AI dashboards auto-generate insights from SQL queries.
Content/SEO Writing	HIGH	GenAI handles bulk content production.
Basic Frontend Dev (HTML/CSS)	MEDIUM	AI code generation covers simple UI work but not complex architecture.
Generalist SWE (no specialization)	MEDIUM-HIGH	AI writes 20-30% of code at Microsoft. Generalist value is declining.
Customer Support (Tier 1)	VERY HIGH	Conversational AI handles 70%+ of tier-1 queries.

2.4 Which Jobs Are GROWING Because of AI

- **LLM / Generative AI Engineer:** AI systems need builders, not just users.
- **MLOps Engineer:** Production AI needs infrastructure engineers to maintain it.
- **AI Product Manager:** Someone needs to own AI product strategy and roadmap.
- **Agentic AI Developer:** Multi-agent systems are new engineering frontier.
- **AI Safety / Red Teaming Engineer:** Regulatory compliance + safety requirements expanding.
- **Data Engineer (AI pipelines):** AI needs clean, reliable data pipelines. Demand up 60% YoY.
- **AI Integration Engineer:** Companies integrating AI into existing systems need specialists.
- **Prompt Engineer (advanced):** Complex enterprise AI workflows require expert prompt design.

SECTION 3: WHAT COMPANIES ACTUALLY EXPECT FROM FRESHERS

Role-by-role breakdown of real expectations, not ideal expectations

AI / LLM Engineer (Fresher)

Core Responsibilities:

- Build and maintain RAG pipelines for enterprise document AI
- Fine-tune open-source LLMs on domain-specific data
- Integrate LLM APIs (OpenAI, Anthropic, Gemini) into product features
- Build evaluation frameworks to measure model output quality
- Deploy models as REST APIs using FastAPI/Flask

Day-to-Day Work:

Writing Python code for LLM pipelines, debugging API integrations, running evaluation tests on model outputs, collaborating with product teams on prompts.

Difficulty	Medium-High	Fresher Hiring?	YES — Strong hiring for this role if you have a solid R
India Openings	500-1000+ in India actively (Naukri/Linked	Prep Time 2026)	4-6 months for a serious fresher
Python Level	Advanced (OOP, async, type hints)	Salary Range	■8-15 LPA fresher Product company: ■12-20 LPA
DSA Level	Medium (LeetCode Easy-Medium, system	Math Level)	Basic-Medium (linear algebra for embeddings, probab

Realistic 6-Month Roadmap:

Python → ML basics → Transformers → LangChain → RAG → Fine-tuning → Deploy

Interview Rounds:

1 OA (DSA), 1-2 Technical rounds (Python + LLM concepts), 1 System Design, 1 HR

Portfolio Requirements:

Projects: 2-3 deployed LLM apps with GitHub + live demo (HuggingFace Spaces/Railway) | Cloud: AWS or GCP basics, Docker mandatory

ML Engineer (Fresher)

Core Responsibilities:

- Build and train ML models for specific product use cases
- Write data pipelines to preprocess training data
- Run experiments, track metrics (MLflow, W&B;)
- Collaborate with data scientists to productionize models
- Basic model serving and API creation

Day-to-Day Work:

Training models, analysing metrics, debugging data pipeline issues, writing evaluation scripts, attending ML team standups.

Difficulty	Medium	Fresher Hiring?	YES — High demand, especially in product companies
India Openings	800-1500+ in India	Prep Time	5-6 months
Python Level	Advanced	Salary Range	■7-12 LPA fresher Product: ■10-18 LPA
DSA Level	Medium (LeetCode Easy-Medium required)	Math Level	Medium (statistics, probability, linear algebra — college level)

Realistic 6-Month Roadmap:

Python → Statistics → Classical ML (sklearn) → Deep Learning → MLOps basics → Deploy

Interview Rounds:

OA (DSA+coding) → 2 Technical (ML + coding) → System Design → HR

Portfolio Requirements:

Projects: 2-3 end-to-end ML projects with deployment, not just notebooks | Cloud: AWS SageMaker or GCP Vertex AI basics preferred

MLOps Engineer (Fresher)

Core Responsibilities:

- Build CI/CD pipelines for ML model deployment
- Monitor models in production for drift and performance
- Manage model registries and versioning
- Container and orchestrate model serving with Docker/K8s
- Write infrastructure-as-code for ML platforms

Day-to-Day Work:

Setting up pipelines, fixing deployment issues, monitoring dashboards, collaborating with ML engineers to productionize their models.

Difficulty	Medium (engineering) + learning curve on ML	Fresher Hiring?	MODERATE — Less common than ML Engineer, but growing
India Openings	300-600 in India	Prep Time	5-7 months (need strong DevOps fundamentals first)
Python Level	Advanced	Salary Range	■8-14 LPA fresher GCC: ■12-20 LPA
DSA Level	Medium	Math Level	Low-Medium (not heavy math, more engineering)

Realistic 6-Month Roadmap:

Python → Linux/Docker → CI/CD → Cloud basics → ML → MLOps tools (MLflow, BentoML)

Interview Rounds:

OA → Technical (MLOps concepts, Docker, K8s) → System Design → HR

Portfolio Requirements:

Projects: ML pipeline project with Docker + GitHub Actions + monitoring | Cloud: AWS/GCP required. Terraform or Kubernetes knowledge valued.

Data Scientist (Fresher)

Core Responsibilities:

- Analyze datasets to extract business insights
- Build predictive models for business KPIs
- Create dashboards and visualizations for stakeholders
- A/B test new features and products
- Work with SQL, Python for data extraction and analysis

Day-to-Day Work:

SQL queries, Python analysis, building plots, presenting findings to business teams, model iteration.

Difficulty	Medium — Most accessible AI-adjacent role	Fresher Hiring?	YES — Most common entry point for AI freshers
India Openings	2000+ in India (highest volume of all AI roles)	Prep Time	3-4 months for motivated fresher
Python Level	Intermediate-Advanced	Salary Range	■5-10 LPA fresher Product: ■8-18 LPA
DSA Level	Low-Medium (often OA includes Python scripting)	Math Level	Medium (statistics is critical — A/B testing, hypothesis testing)

Realistic 6-Month Roadmap:

Python → SQL → Statistics → Data viz → ML → EDA projects → Kaggle

Interview Rounds:

OA (SQL + Python scripting) → Case study → Technical → HR

Portfolio Requirements:

Projects: EDA + predictive modelling projects, Kaggle competitions valued | Cloud: SQL mandatory. Cloud nice-to-have.

Applied AI / Backend+AI Hybrid Engineer

Core Responsibilities:

- Integrate AI models into production backend systems
- Build APIs that serve AI model predictions at scale
- Optimize AI inference for latency and throughput
- Build async job queues for batch AI processing
- Write production-grade Python services with AI components

Day-to-Day Work:

Backend engineering + AI integration. Building microservices, testing APIs, handling async processing of AI workloads.

Difficulty	Medium-High (requires both backend and AI knowledge)	Fresher Hiring?	YES — High demand. Companies need engineers who can bridge the gap
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India Openings	1000-2000+ in India (fastest growing hybrid)	Prep Time	5-6 months
Python Level	Expert (FastAPI, async, production patterns)	Salary Range	■8-16 LPA fresher Product/GCC: ■14-25 LPA
DSA Level	Medium-High (system design is important)	Math Levels	Low-Medium

Realistic 6-Month Roadmap:
Python → FastAPI → Docker → Cloud → ML basics → LangChain → System Design

Interview Rounds:
OA (DSA) → Backend technical → AI integration questions → System Design

Portfolio Requirements:
Projects: Backend API project + AI integration (e.g., chatbot with FastAPI + LangChain) | Cloud: Docker mandatory, AWS/GCP preferred, Redis/queues knowledge helpful

SECTION 4: SKILLS DEMAND ANALYSIS

What companies actually want vs. what YouTubers tell you to learn

4.1 Master Skills Rankings

Based on LinkedIn job postings analysis, Stack Overflow Developer Survey 2025, Lightcast job data (1B+ postings), and NASSCOM India talent reports.

Skill	Category	Industry	Fresher Priority	Difficulty	Verdict
Python	Core	10/10	10/10	3/10	MUST HAVE — No exceptions. 70%+ of AI jobs require Python.
SQL	Core	9/10	9/10	3/10	MUST HAVE — Every data/AI role needs SQL. Often tested in OA.
Machine Learning (Sklearn)	ML	9/10	9/10	5/10	MUST HAVE — Core ML algorithms, model evaluation, scikit-learn.
PyTorch	DL Frameworks	9/10	8/10	6/10	STRONGLY RECOMMENDED — Industry standard for DL. TF is secondary.
LangChain / LlamaIndex	GenAI	8/10	9/10	5/10	HIGH VALUE — Core GenAI engineering. Most LLM roles expect this.
RAG Architecture	GenAI	8/10	9/10	5/10	HIGH VALUE — Enterprise AI backbone. Differentiated skill.
Vector Databases (ChromaDB, Pinecone)	GenAI	7/10	8/10	4/10	HIGH VALUE — Integral to RAG pipelines.
Docker	Deployment	8/10	8/10	4/10	MUST HAVE for AI roles — Model packaging and deployment.
FastAPI	Deployment	8/10	8/10	3/10	HIGH VALUE — Standard for building AI API services.
AWS / GCP / Azure	Cloud	8/10	7/10	5/10	HIGH VALUE — At least one cloud platform needed. AWS most common.
DSA (Data Structures & Algorithms)	CS Fundamentals	7/10	10/10	7/10	MUST HAVE for interviews — LeetCode Medium is minimum bar for product companies.
System Design	CS Fundamentals	7/10	8/10	8/10	IMPORTANT — All senior-adjacent fresher roles test this. Start early.
Deep Learning	ML	8/10	7/10	7/10	HIGH VALUE — Neural networks, transformers architecture knowledge.
NLP / Transformers (HuggingFace)	ML	8/10	8/10	6/10	HIGH VALUE — Essential for LLM and text AI roles.
MLflow / Weights & Biases	MLOps	6/10	7/10	3/10	RECOMMENDED — Experiment tracking shows engineering discipline.
Git / GitHub	Dev Tools	9/10	10/10	2/10	ABSOLUTE MUST — Your portfolio lives here. Clean commits matter.

Skill	Category	Industry	Fresher Priority	Difficulty	Verdict
Linux Basics	Dev Tools	7/10	7/10	3/10	RECOMMENDED — Most production AI runs on Linux.
Kubernetes	MLOps	6/10	4/10	8/10	SKIP FOR NOW — Too complex for fresher focus. Learn Docker first.
TensorFlow	DL Frameworks	6/10	5/10	6/10	SECONDARY — PyTorch is preferred. TF if your target company uses it.
Fine-tuning LLMs (LoRA, QLoRA)	GenAI	8/10	7/10	7/10	HIGH VALUE — Major differentiator. Learn PEFT/LoRA on HuggingFace.
Prompt Engineering	GenAI	6/10	7/10	2/10	GOOD BUT OVERSATURATED at basic level. Advanced prompting still valued.
Statistics & Probability	Math	8/10	8/10	5/10	MUST UNDERSTAND — Hypothesis testing, distributions, Bayesian basics.
Linear Algebra	Math	7/10	6/10	6/10	NEED BASICS — Vectors, matrices, eigenvalues for ML understanding.
CI/CD (GitHub Actions)	DevOps	6/10	7/10	4/10	RECOMMENDED — Shows professional engineering practice.
CUDA	Low-level	5/10	2/10	9/10	SKIP — Only for NVIDIA/deep infrastructure roles.
Data Engineering (Airflow, Spark)	Data	7/10	5/10	7/10	NICE TO HAVE — More valuable for data pipelines at scale.
Distributed Systems	CS Fundamentals	6/10	4/10	9/10	SKIP FOR NOW — Senior-level topic. Focus on fundamentals first.
AI Agents (LangGraph, AutoGen)	GenAI	9/10	8/10	6/10	EMERGING MUST HAVE — Agentic AI is the next frontier. Learn now.

4.2 Overhyped vs. Underrated Skills

■ MOST OVERHYPED	■ MOST UNDERRATED
- Basic Prompt Engineering — Everyone claims it, few do it well. Commoditized at entry level. - TensorFlow — PyTorch won. TF is declining in research and startups. - Keras — Absorbed into TF. Legacy tool. - Blockchain+AI — Almost no real jobs. Marketing buzzword. - Generalist 'Data Science' — Too broad. Specialization beats generalization. - No-code AI tools — Not valued by technical employers. - ChatGPT 'expertise' — Table stakes, not a skill anymore.	- SQL — Tested in almost every data/AI interview. Most students neglect it. - FastAPI — The go-to for production AI APIs. Rarely taught in college. - Git/GitHub hygiene — Clean commits, proper READMEs distinguish serious engineers. - Experiment tracking (MLflow, W&B;) — Shows ML engineering rigor. - Evaluation frameworks — How do you KNOW your AI works? Critical. - Linux command line — Critical for deployment. Almost never taught. - RAG architecture depth — Everyone knows the concept; few can implement it well.

4.3 6-Month Priority Skill Stack

If you have exactly 6 months, here is the order to learn things:

Month 1	Foundation Python (advanced) + SQL + Git + Linux basics + Statistics basics
Month 2	ML Fundamentals Classical ML (sklearn) + Pandas + Data visualization + EDA projects
Month 3	Deep Learning + NLP PyTorch + Transformers + HuggingFace basics + Build text classification project
Month 4	GenAI Engineering LangChain + RAG + Vector DBs + LLM APIs + Build RAG chatbot project
Month 5	Deployment + MLOps FastAPI + Docker + GitHub Actions CI/CD + Deploy 2 projects to cloud
Month 6	Polish + DSA + Interview Prep LeetCode Medium × 60 problems + System design basics + Resume + Mock interviews

SECTION 5: CAREER PATH RECOMMENDATIONS FOR MARCO

Specific guidance based on your profile, timeline, and goals

5.1 Your Current Strengths (Objective Assessment)

- Strong project portfolio: Multi-Agent Code Review System (LangGraph, Claude API, Docker), Multimodal RAG, SQL RL Environment, QuantumSEFS
- CGPA 8.84 — Above average. Won't disqualify you anywhere.
- Leadership credentials: ISOC Chairperson (50+ members), Krithomedh Event Core Member
- Hackathon credibility: Top 10 AIDC (600+ teams), 1st Prize WebHathon 3.0
- Already building in the right stack: LangGraph, RAG, Docker, FastAPI, HuggingFace Spaces
- Meta PyTorch Hackathon participation shows initiative and competitive mindset
- Hyderabad location: Strong GCC and product company presence (Microsoft, Amazon, Google GCCs)

5.2 Your Gaps (Honest Assessment)

- DSA: LeetCode 350+ is the FAANG target. You're at ~90 problems. This is your biggest gap.
- System Design: No evidence of deep system design study. Important for product companies.
- Deployment polish: Projects need production-quality deployment, not just demo videos.
- Open source: No major open source contribution. A PR to LangChain/HuggingFace is valuable.
- Interview experience: Mock interviews are critical. Start soon.

5.3 Top 3 Recommended Career Paths For You

PATH #1 (STRONGEST): AI/LLM Engineer + Agentic AI Specialist

You already have the foundational projects (Multi-Agent Code Review, RAG, LangGraph). This is the hottest domain in 2026 with 280%+ YoY job growth. Your existing work directly positions you for this. You need to polish, deploy, and articulate your projects better, not rebuild from scratch.

Target Companies	Krutrim, Sarvam AI, Postman, Sprinklr, HaptikAI, Zoho AI, Microsoft Azure AI team, Amazon Alexa, Google AI GCC
Expected Salary	■12-20 LPA fresher with your profile ■18-30 LPA if FAANG GCC
Timeline	3-4 months polish + interview prep (you're already 50% there)

Action Items:

- Deploy your Multi-Agent Code Review System with proper README, live demo, architecture diagram
- Deploy your RAG system to HuggingFace Spaces with a clean web UI
- Write 2 technical blog posts explaining your projects (Medium/Substack)
- Contribute one PR to LangChain or LlamaIndex repository
- Solve 150 more LeetCode problems (focus on Medium)

Companies desperately need engineers who can both build AI systems AND integrate them into scalable production backends. Your FastAPI + Docker experience is a strong base. This role has the widest job market and the clearest interview process.

Target Companies	Flipkart, Meesho, Zomato, PhonePe, Razorpay, Freshworks, mid-size product startups
Expected Salary	■10-18 LPA fresher ■15-25 LPA at top Indian product companies
Timeline	4-5 months (strengthen backend + system design)

Action Items:

- Build one production-grade backend service with an integrated AI feature
- Study system design: Grokking System Design or Arpit Bhayani's course
- Learn async Python patterns (asyncio, background tasks)
- Get Docker + GitHub Actions working on ALL your projects
- Apply broadly to Indian unicorns and product companies

PATH #3 (STEADY + ACCESSIBLE): ML Engineer at GCC / Product Company

GCCs (Global Capability Centers) in Hyderabad are creating 4.25-4.5 lakh new jobs in 2026. They hire AI/ML engineers at ■12-25 LPA for freshers. Interview bar is lower than FAANG but salary is strong. Microsoft, Amazon, Google, and JPMC all have major GCCs in Hyderabad.

Target Companies	Microsoft MSIDC, Amazon Hyderabad, Google Hyderabad, JPMC Hyderabad, Deutsche Bank Tech, Goldman Sachs
Expected Salary	■12-22 LPA fresher
Timeline	4-5 months

Action Items:

- Target JPMC's engineering programs (you already prepped for Code for Good)
- Build ML projects with clear business impact metrics
- Learn Pandas proficiency + SQL (tested heavily in GCC interviews)
- Study ML system design basics
- Network on LinkedIn with GCC HR and engineers in Hyderabad

5.4 Paths To AVOID

Do NOT pursue these with a 6-month timeline:

- **AI Research / ML Research Scientist:** Requires PhD or exceptional published papers. 3+ years minimum. Wrong path for placement focus.
- **Reinforcement Learning specialist:** Extremely math-heavy. Few fresher openings in India. Very long ramp time.
- **CUDA / Low-level AI infrastructure:** Requires deep systems programming. Only for NVIDIA-type companies.

- **Pure Data Annotation roles:** Low salary ceiling (■3-5 LPA). No career growth in AI engineering direction.
- **Blockchain + AI:** Almost no real job market. Marketing buzzword.
- **Generic Data Scientist at IT services:** Low ceiling at TCS/Infosys (■6-8 LPA). Same effort, product company pays 2x.

SECTION 6: 6-MONTH EXECUTION PLAN

Week-by-week realistic roadmap for a serious CSE student with existing AI/ML base

Assumption: You are starting this plan now (Month 0 = June 2026). You are in 3rd year with existing Python skills, some ML knowledge, and 3-4 deployed projects. This plan builds on your existing base, not from zero. Adjustments made for Hyderabad location, FAANG and Indian product company targets, and realistic college workload.

MONTH 1: Foundation Audit + DSA Kickstart

Weekly Breakdown:

- **Week 1:** Audit all existing projects. For each: add proper README, architecture diagram, setup instructions. Push clean code to GitHub. Register on LinkedIn, Naukri, Cutshort.
- **Week 2:** DSA: Arrays, Strings, Hashing — solve 20 LeetCode Easy/Medium. Deploy QuantumSEFS or Multi-Agent project to HuggingFace Spaces or Railway.
- **Week 3:** DSA: Two Pointers, Sliding Window, Binary Search — solve 20 problems. Write first technical blog post about Multi-Agent Code Review project.
- **Week 4:** DSA: Linked Lists, Stacks, Queues — solve 20 problems. Update LinkedIn with all projects. Begin applying to internships (Internshala, LinkedIn).

Daily Schedule	Morning: 1-2 LeetCode problems (1 hour). Evening: project work or learning (2-3 hours). Night: Reading (30 min).
Month Goals	40-60 LeetCode problems solved. All GitHub projects polished. LinkedIn profile 100% complete.
What to Skip	Don't start learning new AI frameworks. Audit and deploy what you have first.

Weekly Breakdown:

- **Week 5:** Deep dive into RAG architecture: chunking strategies, embedding models (OpenAI, Cohere, BGE), hybrid search. Upgrade your existing RAG project.
- **Week 6:** Fine-tuning: Learn LoRA/QLoRA with HuggingFace PEFT library. Fine-tune a small model (Phi-2 or Mistral-7B) on a custom dataset.
- **Week 7:** LangGraph deep dive: Build a more complex multi-agent workflow (e.g., research agent that can search + summarize + cite). Document it properly.
- **Week 8:** Evaluation frameworks: Learn LangSmith, RAGAS. Add proper evaluation to your RAG project. DSA: Trees and Graphs — 20 problems.

Daily Schedule	Morning: DSA (1 hr). Afternoon (if possible): AI project work (2-3 hr). Evening: HuggingFace/documentation study.
Month Goals	Fine-tuned model on HuggingFace. Upgraded RAG with evaluation. 80-100 LeetCode problems total.
What to Skip	Don't go deep into TensorFlow. PyTorch + HuggingFace is the path.

MONTH 3: Deployment + Production Engineering

Weekly Breakdown:

- **Week 9:** FastAPI mastery: Build a production-grade AI API with authentication, rate limiting, proper error handling, logging. Docker the entire thing.
- **Week 10:** CI/CD: Set up GitHub Actions for automated testing and deployment on ALL projects. Learn basic Docker Compose for multi-service apps.
- **Week 11:** Cloud deployment: Deploy 2 projects to AWS (EC2/ECS) or GCP (Cloud Run). Get comfortable with environment variables, secrets management.
- **Week 12:** MLflow: Add experiment tracking to your ML projects. Write second technical blog post on RAG architecture. DSA: Dynamic Programming basics — 15 problems.

Daily Schedule	Morning: DSA (1 hr). Project deployment work (2-3 hr). Study cloud docs in evening.
Month Goals	3+ projects deployed on cloud with proper CI/CD. 120 LeetCode problems total.
What to Skip	Don't over-invest in Kubernetes now. Docker is sufficient for fresher interviews.

MONTH 4: System Design + Open Source

Weekly Breakdown:

- **Week 13:** System Design: Study 'Grokking System Design' or Arpit Bhayani's course. Focus on: URL shorteners, rate limiters, notification systems, search engines.
- **Week 14:** ML System Design: How would you design a recommendation system? A fraud detection system? A document AI system? Practice 2-3 end-to-end ML system designs.
- **Week 15:** Open Source: Find a good first issue in LangChain, LlamaIndex, or HuggingFace PEFT. Submit one real PR. Even documentation PRs count.
- **Week 16:** Mock interviews: Schedule 3-5 mock technical interviews (Pramp, Interviewing.io, or with peers). DSA: Greedy, Backtracking — 15 problems. 150 LeetCode total.

Daily Schedule	Morning: DSA + system design (1.5 hr). Afternoon: open source or mock interviews. Evening: system design study.
Month Goals	150+ LeetCode. 1 merged open source PR. System design confident in 5+ scenarios.
What to Skip	Don't build new projects. Polish and deploy existing ones.

MONTH 5: Applications + Interview Rounds

Weekly Breakdown:

- **Week 17:** Resume: Final version of resume with all projects, metrics, deployment links. Have 2 versions: 1-page for startups, 2-page for FAANG. LaTeX preferred.
- **Week 18:** Apply aggressively: 50+ applications. Tier 1 (FAANG/GCC): 10 apps. Tier 2 (Unicorns): 15 apps. Tier 3 (Product startups): 25 apps. Track everything in a spreadsheet.
- **Week 19:** Interview prep: LLM/AI concepts revision (attention mechanisms, RAG internals, fine-tuning theory). Prepare 10 STAR stories for behavioral rounds.
- **Week 20:** Interview rounds: Keep applying while doing interviews. Don't stop for any single company. Every rejection is data.

Daily Schedule	Morning: LeetCode revision (1 hr). Application block (1 hr). Interview prep (2 hr). Evening: Keep learning (30 min).
Month Goals	50-100 applications sent. 5-15 interview calls. 3-5 technical rounds completed.
What to Skip	Don't wait for the 'perfect moment' to apply. Apply now and improve as you go.

MONTH 6: Offer Closing + Continued Learning

Weekly Breakdown:

- **Week 21-22:** Continue interview process. Negotiate offers — most freshers don't negotiate. Even 1-2 LPA negotiation is worth it. Research company comp before interviews.
- **Week 23-24:** If no offer yet: expand application scope (remote startups, smaller product companies), seek referrals, reach out to alumni at target companies on LinkedIn.

Daily Schedule	Interviews + applications (primary focus). Keep 30 min/day of DSA to maintain sharpness.
Month Goals	1-3 offers in hand. Final decision made. Notice period/joining confirmed.
What to Skip	Don't panic if month 6 arrives without an offer. Campus placements + off-campus are parallel tracks.

6.2 Resource Stack — Best Free Resources

DSA	LeetCode (NeetCode 150 list) Striver's A2Z Sheet freeCodeCamp algorithms NeetCode YouTube
ML Fundamentals	fast.ai Practical Deep Learning Andrew Ng's ML Specialization (Coursera) StatQuest YouTube
Deep Learning	Andrej Karpathy's YouTube (neural nets from scratch) PyTorch official tutorials
LLM Engineering	LangChain docs LlamaIndex docs HuggingFace courses (free) Weights & Biases courses
RAG + Agents	LangGraph docs DeepLearning.ai short courses (most free) Anthropic docs
MLOps / Deployment	MLOps.community Made With ML (Goku Mohandas) Docker official docs
System Design	Arpit Bhayani's system design course ByteByteGo book Grokking System Design
Cloud	AWS free tier + AWS Skill Builder Google Cloud Skills Boost (free courses)
Interview Prep	Pramp (free mock interviews) Interviewing.io STAR method guides
Stay Updated	Papers With Code Hugging Face blog AI Twitter/X (Andrej Karpathy, Sebastian Raschka) DeepLearning.ai newsletter

6.3 Realistic Outcome Probability Estimates

Based on your profile (8.84 CGPA, 4 strong AI projects, hackathon wins, Hyderabad location), here are honest probability estimates after 6 months of serious execution:

Outcome	If You Execute Plan	If You Execute 70%	If You Execute 40%
Internship offer (any company)	95%	80%	50%
■6-10 LPA offer	95%	85%	60%
■10-15 LPA offer (product co.)	70%	45%	15%
■15-20 LPA offer (GCC/unicorn)	40%	20%	5%
■20+ LPA offer (FAANG/top co.)	15%	5%	1%
Remote startup role	60%	40%	20%
FAANG interview call	30%	15%	3%

Reality Check: These probabilities assume you send 50-150 applications, do 5+ mock interviews, deploy your projects properly, and stay consistent on DSA. The biggest predictor of job outcomes is not intelligence — it is *the number of high-quality applications submitted* combined with *interview preparation hours*. Most students fail not from lack of skills but from not applying enough.

6.4 GitHub & LinkedIn Strategy

GitHub — What Recruiters Actually Look At:

- Contribution graph: Green squares every day shows consistency. At minimum, commit 5+ days/week.
- README quality: Each project needs: What it does, Why it matters, How to run it, Architecture diagram, Live demo link.
- Pinned repositories: Pin your 6 best projects. Quality > quantity.
- Stars and forks: Rare but valuable signal. Make at least one truly useful tool that others star.
- Commit messages: Write meaningful commit messages. 'Fixed stuff' is a red flag. Use conventional commits.
- No dead repos: Remove or archive projects with zero documentation. They hurt more than help.

LinkedIn — Actual Optimization That Works:

- Headline: NOT 'Student at VNR VJIE'. USE: 'AI/LLM Engineer | Multi-Agent Systems | RAG | LangGraph | Open to opportunities'
- Featured section: Pin your 3 best projects with live links. This is the first thing recruiters click.
- About section: 3-4 sentences. Who you are, what you build, what you're looking for.
- Skills: Add exactly: Python, Machine Learning, LangChain, PyTorch, RAG, Docker, FastAPI, LLM Engineering, NLP, MLOps
- Engage: Comment on posts by Karpathy, Chip Huyen, Sebastian Raschka. Visibility matters.
- Connect: 50+ relevant connections/week. HR managers at target companies. AI engineers. Hyderabad tech community.
- DM strategy: Personalized message + specific project question = 30% response rate. Generic 'Hi I need job' = 1%.

BONUS: INTERVIEW PREPARATION DEEP DIVE

Exact topics to study for AI/ML engineer interviews at product companies

B1. LLM/AI Concept Questions — What Interviewers Actually Ask

Based on interview reports from Glassdoor, LeetCode discuss, and Blind forums (2025-2026):

Transformers & Attention

- Explain the attention mechanism. What is the computational complexity of self-attention?
- What is the difference between encoder-only, decoder-only, and encoder-decoder transformer architectures? Give examples of each.
- What is positional encoding and why is it needed?
- How does multi-head attention work? Why use multiple heads?
- What is the difference between BERT and GPT? When would you use each?

RAG & LLM Engineering

- Walk me through how you would build a RAG system from scratch. What are the key components?
- What is chunking strategy in RAG? How do you choose chunk size?
- What is semantic search? How does it differ from keyword search?
- What are the failure modes of RAG systems? How do you evaluate them?
- What is RAGAS and how do you use it to evaluate RAG quality?
- How would you handle hallucination in an LLM-powered production system?

Fine-tuning

- What is LoRA? How does it reduce the number of trainable parameters?
- What is QLoRA? Why is it used?
- When would you use fine-tuning vs. RAG? Trade-offs?
- What is RLHF? High-level explanation of how it works.
- What data format is needed for instruction fine-tuning? Show an example.

MLOps & Deployment

- How would you deploy an LLM as a production API? Walk through the architecture.
- What is model drift? How do you detect and handle it?
- Explain the CI/CD pipeline for a machine learning project.
- How would you handle model versioning and rollback?
- What is the difference between batch inference and online inference? When to use each?

Agentic AI

- What is an AI agent? How does it differ from a simple LLM API call?
- What is ReAct prompting? How does it enable tool use?
- How does LangGraph differ from LangChain for agent workflows?
- What are the main challenges in building reliable multi-agent systems?

- How would you evaluate whether an AI agent is working correctly?

B2. DSA Topics Priority for AI Roles

Companies like Amazon, Microsoft, Flipkart, and Meesho test DSA even for AI/ML roles. Here is the priority order based on actual interview reports:

MUST KNOW (Highest Frequency)

- Arrays & Strings — Two pointers, sliding window, prefix sums
- HashMap & HashSet — Frequency counting, grouping, two-sum variants
- Binary Search — On arrays and on answer space
- Linked Lists — Reversal, cycle detection, merge sorted lists
- Trees — BFS, DFS, LCA, serialize/deserialize
- Graphs — BFS, DFS, topological sort, shortest path (Dijkstra)
- Dynamic Programming — 0/1 knapsack, LCS, LIS, grid problems

GOOD TO KNOW (Medium Frequency)

- Heaps/Priority Queues — Top K elements, merge K sorted arrays
- Stack & Monotonic Stack — Next greater element, histogram
- Backtracking — Permutations, combinations, N-Queens
- Greedy — Interval scheduling, jump game
- Tries — Word search, autocomplete

NICE TO HAVE (Low Frequency)

- Segment Trees — Range queries
- Union-Find — Connected components
- Bit Manipulation — XOR tricks
- Advanced Graph — Minimum spanning tree, strongly connected components

B3. Resume Checklist for AI Roles

Every item below should be true for your resume before you send it to a product company:

Format

- LaTeX format preferred for technical roles (use Overleaf). Clean, professional look.
- 1 page for startups, 2 pages acceptable for FAANG/product companies with strong projects.
- Consistent formatting: same font, same bullet style, same date format throughout.
- No photos, no colors (unless tasteful). Black and white prints perfectly.
- Contact: email + phone + LinkedIn + GitHub + HuggingFace profile URL.

Projects Section (Most Important)

- Each project: Name | Tech Stack | Impact metric | GitHub link | Live demo link
- Quantify impact: 'Reduced inference latency by 40%' not 'Made model faster'
- Explain what problem it solves, not just what it does technically
- Order projects by relevance to the role, not chronologically
- 3-4 projects is ideal. More dilutes. Less is too little.

Skills Section

- List skills under categories: Languages | ML Frameworks | GenAI Tools | DevOps | Cloud
- Only list skills you can be interviewed on. Listing 'Kubernetes' when you've only watched a tutorial = danger.
- Put the most relevant skills first for each category.

Experience / Leadership

- Quantify everything: '50+ members', '600+ participants', 'Top 10 of 600+ teams'
- Frame leadership as impact: 'Organized X, resulting in Y'
- Include hackathon wins with scale context (how many teams competed)

ATS Optimization

- Use exact skill keywords from the job description (AI, LLM, RAG, FastAPI etc.)
- Avoid tables, text boxes, and complex formatting — ATS often fails to parse them
- Standard section headings: Education, Experience, Projects, Skills

BONUS: INDIA AI ECOSYSTEM MAP

Key players, cities, and opportunities in the Indian AI job market

B4. India AI Company Tier Map

TIER 1 — FAANG GCCs in India

■25-90 LPA fresher
(IIT/NIT grade)

Google India (Bangalore/Hyderabad), Microsoft MSIDC (Hyderabad), Amazon India (Bangalore/Hyderabad), Meta India, Apple India. Selection rate: <5%. Requires top DSA + ML fundamentals + often research background. For most students, FAANG is a 2-3 year goal after starting at Tier 2.

TIER 2 — Indian Unicorns & Major Product Cos

■15-40 LPA fresher

Flipkart, Meesho, Zomato, Swiggy, PhonePe, Razorpay, CRED, Zepto, Groww, Zerodha. Selection rate: 10-20%. DSA Medium + solid AI project portfolio. Most realistic 'stretch goal' for prepared freshers. These companies build real AI at scale and offer strong engineering culture.

TIER 3 — GCCs of Global Cos

■12-25 LPA fresher

JPMC Tech India, Goldman Sachs, Deutsche Bank Tech, Walmart Global Tech, Wells Fargo, SAP India, Salesforce India, Adobe India. Hyderabad has 900+ GCC units. Stable, well-paying, good work-life balance. Strong target for Hyderabad students.

TIER 4 — AI Startups

■8-20 LPA fresher (equity upside)

Sarvam AI, Krutrim, Ola Krutrim, Postman, HaptikAI, Glean, Yellow.ai, Uniphore, Sprinklr, Observe.AI, BrowserStack AI. High learning curve, direct impact, equity potential. Best place to get production experience fast. Often don't require strong DSA.

TIER 5 — IT Services AI Roles

■6-10 LPA fresher

TCS AI/ML division, Infosys Nia, Wipro Holmes, Accenture AI, Cognizant AI. CAUTION: Salary ceiling is low. Growth is slow. If you have AI skills, aim higher. Only fallback option — don't target these if you're executing this plan seriously.

B5. Hyderabad-Specific Opportunities

You are in Hyderabad — one of India's top 2 AI hiring cities. This is a genuine advantage. Hyderabad has 25% of India's AI job openings (12,500+ roles) with salaries of ■15-70 LPA for mid-senior roles. The cost of living is 30-35% lower than Bangalore — meaning ■15 LPA in Hyderabad has similar purchasing power to ■20 LPA in Bangalore.

Company	AI Focus	Hiring Volume	Fresher Notes
Microsoft MSIDC	Azure AI, Copilot, GitHub	Very High	Strong GCC fresher hiring
Amazon Hyderabad	Alexa, AWS AI, Robotics	Very High	Leadership Principles + DSA focus
Google Hyderabad	Search AI, Cloud AI, YouTube AI	High (selective)	Requires top-tier DSA

Company	AI Focus	Hiring Volume	Fresher Notes
JPMC Tech India	Fintech AI, Fraud Detection	High	Strong fresher programs incl. Code for Good
Apple India	Siri, AI/ML platform	Medium (very selective)	Systems programming + ML
Goldman Sachs	Trading AI, Risk ML	Medium	Quant + ML focus
Deutsche Bank Tech	Risk Models, NLP for docs	Medium	Finance domain knowledge helpful
Walmart Global Tech	Retail AI, Demand Forecasting	Medium-High	Good fresher pipeline
Qualcomm India	Edge AI, Mobile AI	Medium	Systems + AI combination
Sarvam AI	Indic LLMs	High for right profile	Startup culture, high learning velocity

B6. 30-Day Quick Start — If You Start Today

Can't wait for 6 months? Here's what to do in the next 30 days to immediately increase your chances:

Day 1-3	Audit ALL GitHub repos. Add README to every project. Deploy at least 1 project to a live URL.
Day 4-7	Rebuild LinkedIn headline and About section. Add all projects to Featured section. Connect with 30 Hyderabad tech people.
Day 8-14	Solve 20 LeetCode problems (NeetCode Easy list). One problem per day minimum. Track in spreadsheet.
Day 15-21	Apply to 20 companies: 5 startups (no DSA gate typically), 10 mid-tier product companies, 5 GCCs. Use LinkedIn Easy Apply + direct career pages.
Day 22-28	Write one technical blog post on Medium about your Multi-Agent Code Review project. Share on LinkedIn. This single action increases recruiter inbound 3-5x.
Day 29-30	Schedule 2 mock interviews (Pramp is free). Record yourself answering 'Tell me about your projects.' Watch it back. Painful but necessary.

Final Word: The AI job market in 2026 is the best it has ever been for freshers with real skills. The talent supply-demand gap works entirely in your favor. Companies need people who can build and deploy real AI systems — not people who have taken every online course but never shipped anything. Your 4 existing projects (Multi-Agent, RAG, SQL RL, QuantumSEFS) are more valuable than they appear. Polish them, deploy them, document them, and talk about them confidently. The difference between **■8 LPA** and **■18 LPA** for a fresher is almost entirely about *how well you communicate what you've built*, not how many additional frameworks you know. Execute the plan. Apply aggressively. Negotiate everything.

APPENDIX: REFERENCES & SOURCES

All data sources used in this report — verified and cited

Primary Data Sources

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- Made With ML — MLOps course by Goku Mohandas (madewithml.com)
- NeetCode — DSA patterns and LeetCode prep (neetcode.io)
- LangChain documentation (langchain.com)
- Papers With Code — Research benchmarks (paperswithcode.com)
- Arpit Bhayani — System design courses (arpitbhayani.me)

Data Reliability Notes

- Salary figures aggregated from multiple sources. India salary data has $\pm 15\text{-}20\%$ variance by company tier, city, and negotiation.
- Layoff statistics from Crunchbase and Layoffs.fyi updated weekly. Figures reflect data as of May 2026.
- Job growth percentages (280% agentic AI, 135.8% prompt engineering) from specialized analytics platforms — directional trends, not precise measurements.
- Fresher hiring probabilities are author estimates based on aggregated market data patterns, not statistically validated predictions.
- Company AI strategies sourced from public announcements, news, and career pages as of May 2026.

Disclaimer: This report is for educational and career guidance purposes. The AI job market changes rapidly. All statistics are from publicly available sources as of May 2026. Always cross-reference with current job postings and LinkedIn data for up-to-date information. The author has made best-effort attempts to verify all claims but cannot guarantee complete accuracy of third-party data.

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